

SMART INDIA HACKATHON 2026

CALXION

Team submission · Ministry of Coal problem statement

Problem Statement ID — SIH26025

Problem Statement Title — AI-Enabled Low-Cost Real-Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

Theme — Disaster Management

PS Category — Hardware

Team ID — To be allotted on portal

Team Name (Registered on portal) — CALXION



Underground Sensor Nodes
Geophone · Tilt · Extensometer



LoRaWAN Through-Earth Link
868/915 MHz mesh to surface



Edge + Cloud AI Engine
LSTM + XGBoost risk scoring



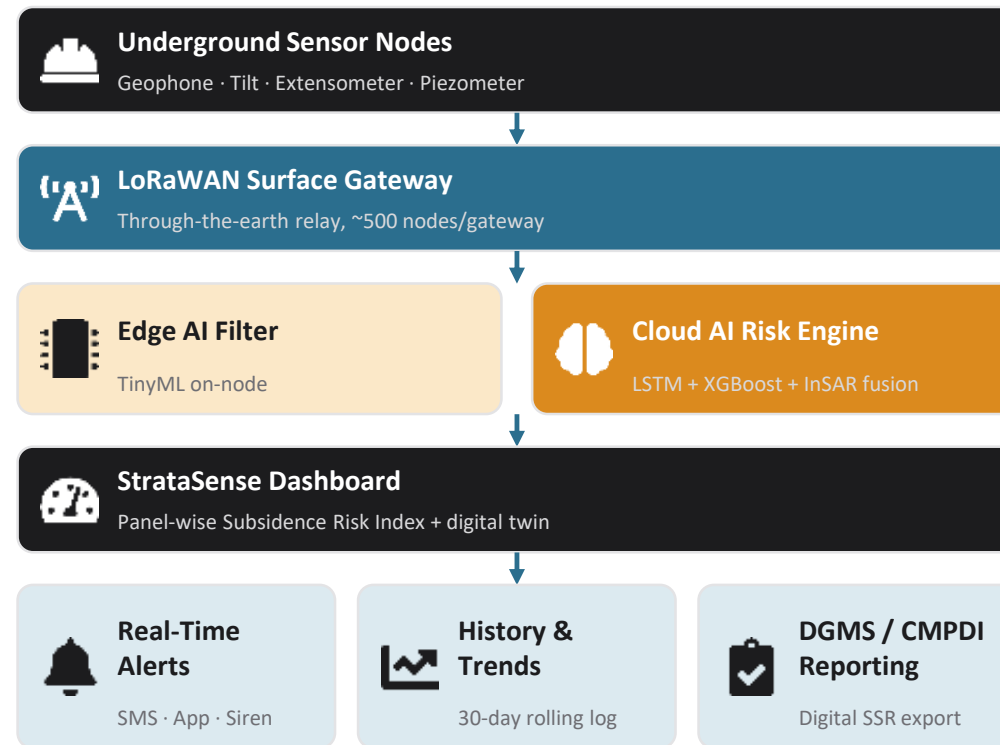
Real-Time Early Warning
SMS · App · Mine-Wide Siren

StrataSense AI

AI-Enabled, Low-Cost Subsidence Monitoring for Underground Coal Mines

Proposed Solution

- **Sensor-to-Dashboard Ecosystem:** a low-cost network of DGMS/CIMFR-certifiable, intrinsically-safe underground sensor nodes paired with a live web + mobile risk dashboard.
- **Real-Time Strata Monitoring:** MEMS tilt, tri-axial geophone and borehole-extensometer nodes stream ground movement and micro-tremor data every few seconds.
- **AI Subsidence Risk Engine:** an LSTM + XGBoost ensemble fuses live sensor readings with mine void maps and geology to output a panel-wise Subsidence Risk Index.
- **Satellite Fusion Layer:** periodic Sentinel-1 InSAR displacement maps calibrate the ground network, bridging second-by-second sensing with wide-area surface trends.
- **Digital Twin & Void Map:** a 3D model of worked-out panels overlays live readings so engineers see exactly which zone is destabilising, not just one score.
- **Edge-AI False-Alarm Filter:** on-node inference tells blasting vibration apart from genuine strata failure before an alert ever leaves the mine.
- **Multi-Channel Early Warning:** automated SMS, app push and mine-wide siren/beacon alerts in Hindi and regional languages to miners, management and DGMS-facing dashboards.

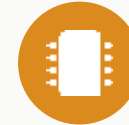


End-to-end pipeline: sensing → through-earth comms → edge + cloud AI → dashboard & alerts

TECHNICAL APPROACH

Hardware & Technology Stack

- **Sensing Hardware:** MEMS tilt/accelerometer, tri-axial geophones (microseismic), vibrating-wire borehole extensometers & piezometers, stress cells, solar + battery power, Ex-ia / flameproof enclosure (IS/IEC 60079).
- **Communication:** LoRa SX1276 through-the-earth transceiver (868/915 MHz), LoRaWAN star-of-stars mesh, 4G / leaky-feeder gateway backhaul to surface.
- **Firmware & Edge AI:** ESP32 / STM32 low-power MCU, FreeRTOS, TensorFlow Lite Micro (TinyML) for on-node false-alarm filtering, OTA updates.
- **Backend:** FastAPI, Python, PostgreSQL + TimescaleDB, MQTT broker, Redis, Celery, Docker.
- **AI / ML:** PyTorch, LSTM/GRU time-series forecasting, XGBoost & LightGBM risk scoring, scikit-learn, ONNX Runtime for edge export.
- **Geospatial & Satellite:** Sentinel-1 SAR, InSAR/SBAS processing (SNAP), QGIS, GDAL, digital elevation & void-map overlay.
- **Frontend & Deployment:** React.js, React Native, Mapbox GL, Chart.js, Socket.io, Kubernetes, AWS / on-prem hybrid, Grafana, Prometheus.



Edge AI / TinyML



LoRaWAN Mesh



Sentinel-1 InSAR



TimescaleDB



PyTorch · XGBoost



Digital Twin / QGIS



Kubernetes · AWS



Ex-ia / IEC 60079



React Native App

FEASIBILITY AND VIABILITY



Feasibility

1. **Technical:** geophones, extensometers and LoRa/TinyML are mature, field-proven technologies already used in Indian mines.
2. **Economic:** solar + LoRa removes trenching and cabling, the single largest cost in underground instrumentation.
3. **Operational:** needs DGMS / CIMFR flameproof approval and integration with existing Support Stability Report workflows.



Viability

4. **Market opportunity:** 350+ underground coal mines under Coal India alone; strata failure is a leading cause of underground fatalities.
5. **Sustainability:** sensor nodes scale panel-by-panel; the same AI core extends to other Ministry of Coal safety problem statements.



Challenges

6. **Explosive-atmosphere compliance:** every board must be Ex-ia or flameproof per IS/IEC 60079, unlike open-pit hardware.
7. **Underground RF propagation:** LoRa attenuates through coal and rock; gateway spacing must match seam geometry.
8. **Sparse failure data:** labelled subsidence datasets are limited, so the model leans on rock-mechanics features.



Use Cases

9. **Evacuation triggers:** automatic alerts when a panel crosses a critical risk threshold.
10. **Digital SSR:** sensor-backed reports replacing paper logs for DGMS inspection.
11. **Land use & compensation:** rehabilitation maps for old workings, plus objective logs for faster damage-claim resolution.



Business Potential

1. **AMC / subscription:** per-panel monitoring contracts with CIL subsidiaries (ECL, BCCL, CCL, SECL).
2. **Tiered hardware bundles:** basic tilt-only nodes up to full multi-sensor panel kits.
3. **Certification partnerships:** bundled DGMS / CIMFR type-approval support for faster adoption.
4. **Scheme alignment:** eligible under Ministry of Coal S&T and mine-safety funding.



Engineering Solutions

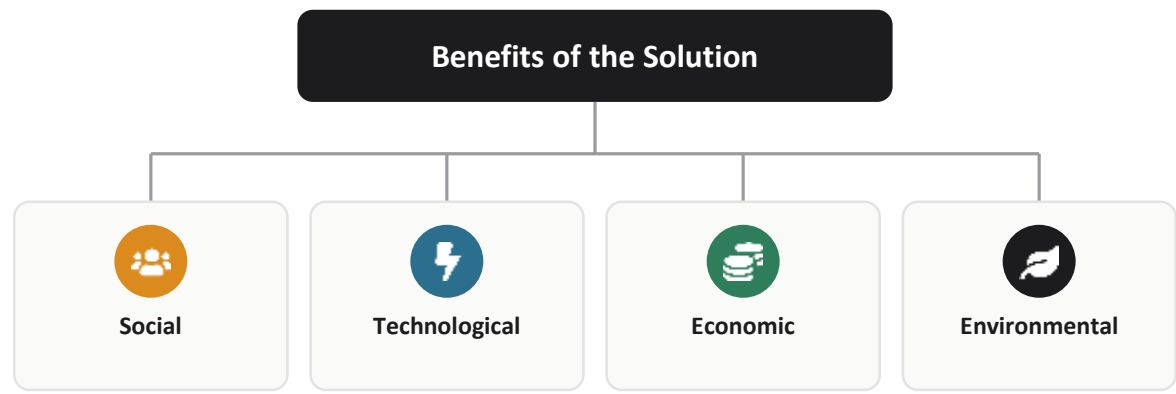
5. **Ex-ia-first design:** every energy-storing element stays under IS safety limits from day one.
6. **Adaptive mesh relay:** self-healing LoRa mesh, hop count tuned per seam.
7. **Physics-informed ML:** extraction ratio and depth of cover cut the labelled-data need.



Supporting Facts for Feasibility and Viability

- Roof and strata failure has historically been among the leading causes of fatalities in Indian underground coal mines, per DGMS accident records.
- Peer-reviewed reviews report ML models such as XGBoost reaching over 94% accuracy in mining-induced subsidence prediction when fused with InSAR, GNSS and UAV data.

IMPACT AND BENEFITS



Potential Impact on the Target Audience

- **Mine Management & CIL Subsidiaries:** prevents production stoppages and equipment loss from unplanned panel collapse.
- **Miners & Support Staff:** an early-warning layer for the single largest historical cause of underground fatalities.
- **DGMS & Regulators:** a transparent, sensor-backed record replacing periodic manual strata observation.
- **Surface Communities:** advance warning of subsidence beneath villages and roads above old workings, as seen in Jharia and Raniganj.



Social

- Eases anxiety of working under uncertain roof and strata conditions.
- Protects surface villages long affected by subsidence, e.g. Jharia and Raniganj.
- Builds trust through transparent, sensor-backed risk data.



Technological

- Brings low-cost edge-AI sensing to a segment still using manual readings.
- Complements CIL's satellite InSAR monitoring with continuous ground truth.
- A reusable sensor-to-AI pipeline, extensible to other strata-control PSs.



Economic

- Targets fewer unplanned panel-downtime days from undetected movement.
- Aims for hours-to-days of warning versus today's reactive surveys.
- Removes underground cabling cost versus fibre-optic systems.



Environmental

- Supports precise extraction planning, limiting over-extraction subsidence.
- Feeds into post-mining land rehabilitation and afforestation planning.
- Cuts the footprint of reactive, ad-hoc emergency responses.

RESEARCH AND REFERENCES

Research Keywords

Mine subsidence prediction

Underground coal mine safety

Microseismic monitoring

LoRaWAN through-the-earth

InSAR/SBAS deformation mapping

Strata movement early warning

Intrinsically safe IoT sensors

Digital twin void mapping

XGBoost & LSTM risk scoring

Coalfield subsidence (India)

Low-cost geotechnical sensors

References — Research & Best Practices

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- IoT LoRaWAN Wireless Sensor Network for Underground Mine Monitoring, MDPI mdpi.com/1424-8220/24/21/6971
- Land Subsidence Detection in Jharia Coalfield — DInSAR, GPS & Levelling ias.ac.in/public/Volumes/jess/124/06/1359-1376.pdf
- Surface Deformation Monitoring of Raniganj Coalfield — InSAR & DGPS tandfonline.com/doi/full/10.1080/19475705.2024.2375546
- Directorate General of Mines Safety (DGMS) — official regulator dgms.gov.in
- CIMFR — Flameproof & Intrinsically Safe Equipment Testing cimfr.res.in/division/flame-proof-and-equipment-safety.html

Existing Initiatives We Build On

- Coal India × NRSC/ISRO satellite dashboard MoU (Jul 2026) — satellite-only, periodic revisit isro.gov.in/NRSC_ISRO_MoU_BCCL_CMPDI.html
- Worldsensing — commercial GNSS/tiltmeter ground-deformation monitoring for mining worldsensing.com/mining

Comparison with Existing Approaches

Capability	StrataSense AI	Manual Reading	Satellite InSAR-Only	Cabled DFOS
Real-time continuous alerts	Yes	No	No	Yes
Underground ground-truth sensing	Yes	Yes	No	Yes
No underground cabling needed	Yes	Yes	Yes	No
AI prediction engine	Yes	No	Partial	Partial
DGMS / Ex-ia compliance built-in	Yes	N/A	N/A	Partial
Wide-area satellite fusion	Yes	No	Yes	No
Panel-level digital twin	Yes	No	No	Partial
Multilingual field alerts	Yes	No	No	No

StrataSense AI targets the gap between today’s manual instrument rounds and satellite-only monitoring: continuous, low-cost, panel-level ground truth fused with wide-area InSAR.